Decision context

Joint planner

Continuation state selection (only for editing)

Action selection

submit

Submit if
• Budget exhausted
• All constraints satisfied
• Planner selects to stop

Final output GM-best

Retrieved guidance

Query skill

Generate / edit

Budget -1

Execution & Verifier

State Update

Interaction History

(a)

Next decision

Decision context

Original Prompt

four sparkling trumpets behind five horses to the right of four purple backpacks

Atomic constraints

× 4 Sparkling trumpets

× 5 Horses

× 4 Purple backpacks

Behind × 1

Right of × 1

Remaining budgets 2/5

Image candidates (verified)

I0 GM=0.787 I1 GM=0.269 I2 GM=0.222

Retrieved guidance

counting spatial attribute ...

Joint Planner

Continuation state selection

I0 GM=0.787 I1 GM=0.269 I2 GM=0.222

Action Selection

Generate Edit Query skill Submit

Edit plan

One executable action

Source image: I1.
Target: backpack 5 -> 4.
remove: one extra backpack.
Preserve: 5 horses, 4 sparkling trumpets, purple color, and spatial relations. ...

GM-best
Latest
Selected

Image Execution & Verification

Editor

I1

Edit instruction

New candidate I3

verified results

objects

relations

Behind Right of

All passed
I3 GM=0.998

State Update

Append I3

Update GM-best

Update budget

Record action feedback

History retained & consumed budget never reset

Interaction History

I0 GM=0.787 I1 GM=0.269 I2 GM=0.222 I3 GM=0.998

update

(b)

Next decision